Ann prac5

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import Perceptron

# Training data
X = np.array([
   [1,1], [2,2], [3,3],
   [1,3], [2,4], [3,5]
])

y = np.array([0, 0, 0, 1, 1, 1])

# Train perceptron
model = Perceptron(max_iter=1000, eta0=0.1)
model.fit(X, y)

# Create decision region
xx, yy = np.meshgrid(
   np.linspace(0, 4, 200),
   np.linspace(0, 6, 200)
)

Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

# Plot
plt.contourf(xx, yy, Z, alpha=0.3)
plt.scatter(X[:,0], X[:,1], c=y, edgecolor='k')

plt.xlabel("Feature 1")
```

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plt.ylabel("Feature 2")
plt.title("Perceptron Decision Region")

plt.show()
```